

Practical Assessment: Hand Gesture Recognition using Teachable Machine**1. Introduction**

Classes: Thumbs Up, Peace Sign, Open Palm

Purpose: Train a machine learning model using Teachable Machine to recognise hand gestures, demonstrating supervised learning and the importance of dataset quality.

2. Data Collection Process

Procedure:

Collected approximately 110 images per class using the webcam, Images varied by distance, position (left, right, centre), rotation, and lighting and Demonstrators face was visible in all images.

Challenges:

- A major challenge was that smaller datasets made it harder for the model to correctly identify hand gestures.
- Some gestures, particularly Peace Sign and Open Palm, were visually similar, leading to occasional misclassifications.
- Variations in lighting, camera angle, and hand rotation also affected model accuracy, requiring careful positioning during data collection.

Observations:

- Increasing the number of images per class significantly improved model performance, showing that more data helps the AI generalise better.
- Thumbs Up was reliably recognised, while Peace Sign and Open Palm were occasionally confused with rotated or close fingers.
- Adding variation in hand position, distance, and lighting helped the model perform more reliably under different conditions.
- Overall, a balanced and diverse dataset was critical for achieving stable and accurate predictions across all three gesture classes.

3. Teachable Machine Experience

Ease of Use:

- User-friendly, no coding was required, easy setup of classes and training.
- Real-time webcam preview helped capture gestures accurately.

Helpful Features:

- Confidence percentages in real-time and quick class management.

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Limitations:

Performance depends on lighting, camera angle and small datasets may cause overfitting.

4. Testing and Evaluation

Results Table:

Gesture	Confidence (original)	Confidence (newconditions)	Notes
Thumbs Up	92–95%	88–90%	Very stable; consistently recognised under different lighting and angles.
Peace Sign	20–85%	65–75%	Initially confused with Thumbs Up when dataset was small; performance improved after adding more images. Slight rotation still affects accuracy.
Open Palm	80–90%	75–85%	Occasionally confused with Peace Sign, especially under dim lighting or rotated fingers; improved after dataset expansion.

5. Learnings and Applications

Learnings:

- Data quality and diversity are critical for accurate predictions.
- Small datasets may overfit; larger and varied datasets improve generalisation.
- Testing under different conditions highlights model strengths and weaknesses.

Applications:

- Professional: Sign language recognition, accessibility tools, smart classroom applications.

6. Conclusion

The project demonstrated that AI models require sufficient and diverse data to accurately recognise gestures. Smaller datasets caused misclassifications, while additional varied images improved the model's confidence and reliability. Overall, the project highlighted the importance of data collection, dataset balance, and variation, showing that even a beginner-friendly tool can produce a functional gesture recognition system.

Figure 1 – Data collection for Thumbs Up class

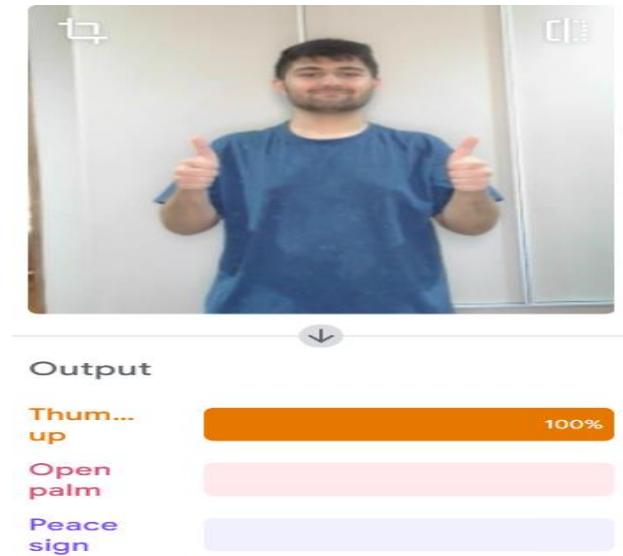


Figure 2 – Class setup screenshot

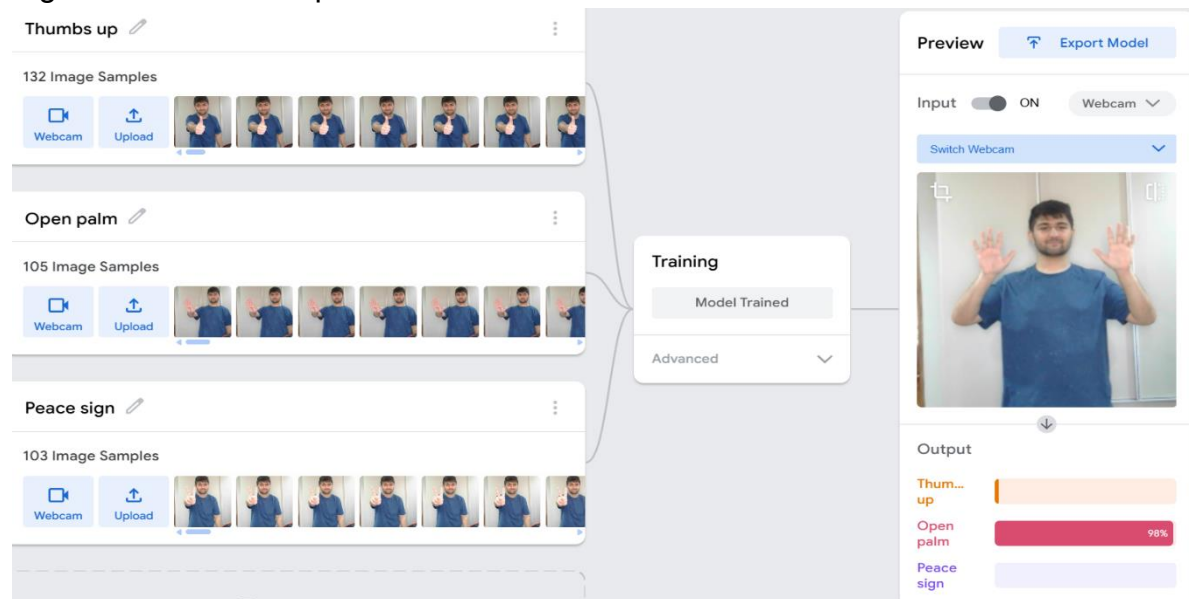


Figure 3 – Training screenshot

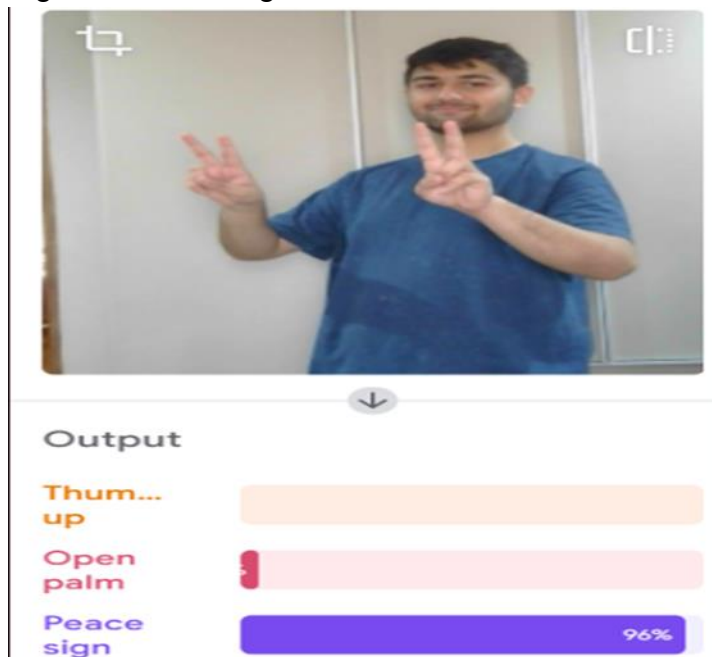


Figure 4 – Testing predictions

